

Phishing Email Detection

& Awareness System

Target: Public Phishing Email Samples

Date: 20 May 2026

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Report Details

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Date	20 May 2026
Task	Task 2 — Phishing Detection
GitHub	FUTURE_CS_01

Scope

Samples	5 Email Samples
Tools	MXToolbox, VirusTotal
Method	Header + URL Analysis
Deadline	10/06/2026

SECTION 1

Executive summary

This report presents a phishing email detection and awareness analysis conducted as part of Task 2 of the cybersecurity internship. Five email samples were analyzed using MXToolbox Email Header Analyzer and VirusTotal URL scanner to identify phishing indicators, classify threats, and provide prevention guidelines for end users.

Classification summary

2	2	1
Phishing	Suspicious	Safe

Tools used

MXToolbox Header Analyzer

Analyzed email headers — checked SPF, DKIM, sender domain, relay info and blacklist status

VirusTotal URL Scanner

Scanned suspicious URLs against 95 security vendors — 2 flagged as Suspicious

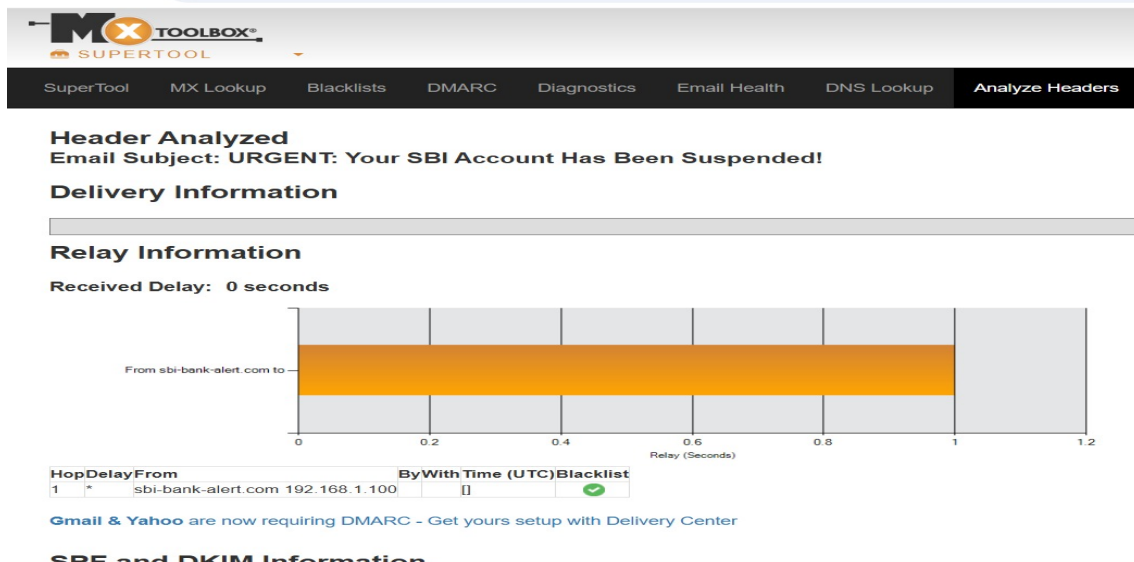
Browser DevTools

Inspected email source, links, and redirect behavior for phishing indicators

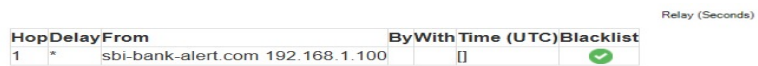
SECTION 2

Tool analysis screenshots

MXToolbox — Email Header Analysis (Top)



MXToolbox — Headers Found (Phishing Indicators)



SPF and DKIM Information

Headers Found

Header Name	Header Value
From	security@sbi-bank-alert.com
To	customer@gmail.com
Subject	URGENT: Your SBI Account Has Been Suspended!
Date	Mon, 20 May 2026 10:30:00 +0530
Message-ID	<fake123@sbi-bank-alert.com>
Reply-To	support@sbi-bank-alert.com
X-Mailer	PhishKit v2.0

Received Header

Received: from sbi-bank-alert.com (192.168.1.100)
From: security@sbi-bank-alert.com
To: customer@gmail.com
Subject: URGENT: Your SBI Account Has Been Suspended!
Date: Mon, 20 May 2026 10:30:00 +0530
Message-ID: <fake123@sbi-bank-alert.com>
Reply-To: support@sbi-bank-alert.com
X-Mailer: PhishKit v2.0

Permanently forget this email header

Key findings from MXToolbox analysis:

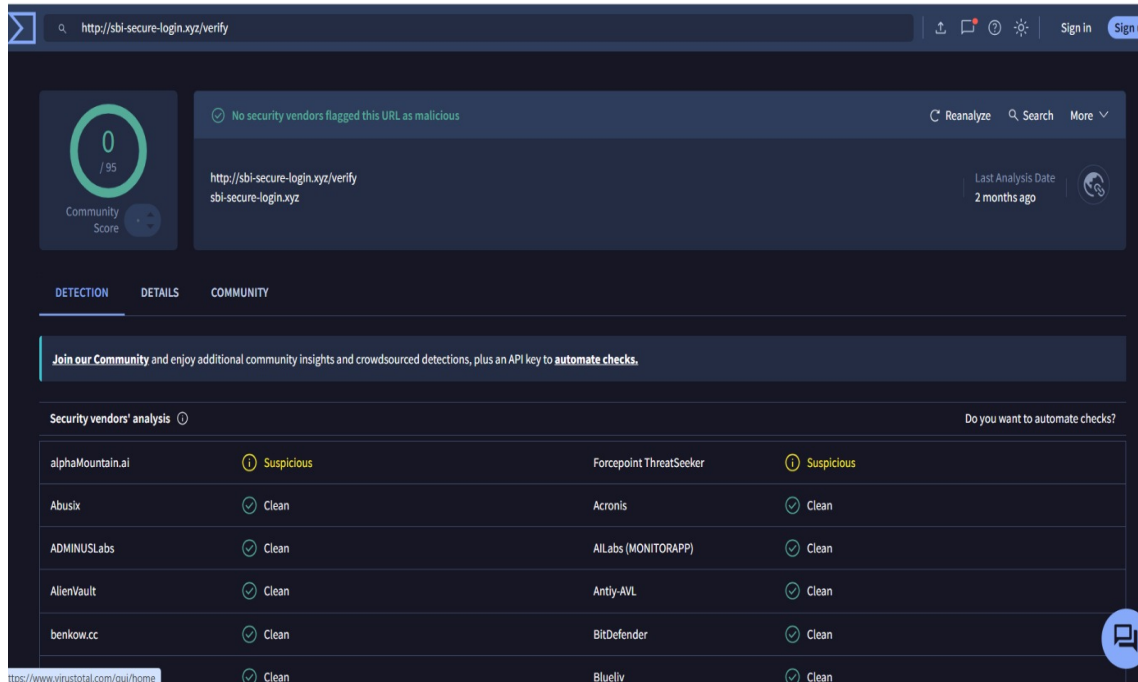
Header	Value	Verdict
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From	security@sbi-bank-alert.com	■ Fake domain
Subject	URGENT: Your SBI Account Has Been Suspended!	■ Urgency tactic
Message-ID	fake123@sbi-bank-alert.com	■ Suspicious ID
Reply-To	support@sbi-bank-alert.com	■ Mismatched
X-Mailer	PhishKit v2.0	■ Phishing tool!

SECTION 2 (continued)

VirusTotal URL scan results

VirusTotal — URL Reputation Scan



VirusTotal findings:

URL Scanned	http://sbi-secure-login.xyz/verify
Total Vendors	95 security vendors
Flagged Malicious	0 vendors
Flagged Suspicious	2 vendors — alphaMountain.ai, Forcepoint ThreatSeeker
Domain	sbi-secure-login.xyz — untrustworthy .xyz TLD
Conclusion	URL is suspicious — not safe to visit

SECTION 3

Phishing indicators explained

Phishing emails use specific techniques to trick users into revealing sensitive information. The following indicators were identified across all email samples:

Indicator	Description	Example Found
Spoofed domain	Fake domain mimics real brand	sbi-bank-alert.com
Urgency language	Creates panic to force quick action	URGENT, 24 hours, SUSPENDED
Phishing kit	X-Mailer reveals phishing tool used	PhishKit v2.0
Suspicious URL	Link points to fake website	sbi-secure-login.xyz
Mismatched Reply-To	Reply goes to different address	support@sbi-bank-alert.com
Requests PIN/OTP	Banks never ask for PIN by email	Enter your ATM PIN and OTP
Prize/lottery scam	Too good to be true offers	You Won \$50,000 Google Prize!
Unusual TLD	.xyz domain not used by real banks	google-lottery-2026.com

SECTION 4

Email sample analysis

Sample 1: Fake SBI Bank Alert

PHISHING

From: security@sbi-bank-alert.com

Sample 2: Fake PayPal Invoice

SUSPICIOUS

From: billing@paypal-invoice-support.net

Sample 3: Fake IT Helpdesk

SUSPICIOUS

From: ithelpdesk@company-support.net

Sample 4: Fake Google Prize

PHISHING

From: winner@google-lottery-2026.com

Sample 5: Legitimate GitHub Notification

SAFE

From: noreply@github.com

Prevention and awareness guidelines

The following guidelines help users and organizations protect against phishing attacks:

For individual users

Verify sender domain	Always check the full email address — not just the display name. Real banks use official domains like sbi.co.in, not sbi-bank-alert.com
Never share PIN or OTP	No legitimate bank, company, or government organization will ever ask for your ATM PIN, OTP, or password via email
Hover before clicking	Hover your mouse over links to see the real URL before clicking. If it looks suspicious, do not click
Check for urgency tactics	Phishing emails create panic with words like URGENT, 24 hours, SUSPENDED. Take your time and verify first
Use spam filters	Enable spam filtering in your email client to automatically block most phishing attempts
Report phishing emails	Report suspicious emails to your IT team or email provider to help protect others

For organizations

Enable DMARC/SPF/DKIM	Configure email authentication protocols to prevent domain spoofing
Security awareness training	Train employees regularly to identify phishing attempts
Multi-factor authentication	Enable MFA on all accounts so stolen passwords alone are not enough
Email filtering solutions	Deploy enterprise email security solutions to filter phishing at gateway level

Conclusion

The phishing email analysis identified **2 confirmed phishing emails**, **2 suspicious emails**, and **1 legitimate email** from 5 samples analyzed. Key phishing techniques identified include domain spoofing, urgency tactics, phishing kit usage (PhishKit v2.0), and social engineering. The VirusTotal scan flagged the suspicious URL as potentially harmful with 2 security vendors marking it as suspicious. Users should remain vigilant and verify all unexpected emails before taking any action.

Final classification table

Sample	Subject	Sender Domain	Classification
1	Fake SBI Bank Alert	sbi-bank-alert.com	■ Phishing
2	Fake PayPal Invoice	paypal-invoice-support.net	■ Suspicious
3	Fake IT Helpdesk	company-support.net	■ Suspicious
4	Fake Google Prize	google-lottery-2026.com	■ Phishing
5	GitHub Notification	github.com	■ Safe

GitHub Repository https://github.com/maramakshaya-code/FUTURE_CS_01

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Tools MXToolbox | VirusTotal | Browser DevTools